Automating Data Import and Relationship Mapping Using Import Sets & Dot Walking

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1. INTRODUCTION

1.1 Project Overview

In modern organizations, data plays a crucial role in decision-making and operational efficiency. However, handling large volumes of data manually often leads to errors, redundancy, and inefficiencies. ServiceNow, being a powerful cloud-based platform, provides tools like Import Sets and Transform Maps to streamline data integration processes.

This project focuses on automating data import workflows and establishing relationships between different tables using Dot Walking. Import Sets allow external data to be temporarily stored and processed before inserting into target tables. Transform Maps help define how incoming data should be mapped, while Dot Walking enables navigation through reference fields to fetch related data.

The system designed in this project ensures smooth data migration, minimizes manual intervention, and maintains relational integrity across tables such as Users, Departments, and Assets.

1.2 Purpose

The main purpose of this project is to:

- **Eliminate manual data entry processes**
- **Automate bulk data import from external sources**
- **Establish relational mapping between tables**
- **Improve data accuracy and consistency**
- **Demonstrate real-time application of ServiceNow features**

Additionally, the project aims to enhance understanding of enterprise-level data handling and automation techniques.

2. IDEATION PHASE

2.1 Problem Statement

Organizations frequently need to import large datasets such as employee records, asset details, or customer information into ServiceNow. Manual data entry is not only timeconsuming but also prone to human errors like duplication, incorrect relationships, and missing data.

There is a need for an automated system that:

- Handles large data efficiently
- Maps relationships correctly
- Reduces redundancy
- Ensures data integrity

2.2 Empathy Map Canvas

Users: System Administrators, Data Analysts

- **Think:** "There must be an easier way to import and manage data."
- **Feel:** Frustrated due to repetitive manual tasks
- **Say:** "Manual entry is slow and error-prone."
- **Do:** Use Excel sheets and manually upload data **Pain Points:**
- Data duplication
- Incorrect relationships
- Time-consuming process **Gains:**
- Automation
- Faster processing
- Accurate data relationships

2.3 Brainstorming

Different approaches were considered:

- Manual data entry → Rejected due to inefficiency
- Script-based import → Complex for beginners
- Import Sets + Transform Maps → Selected solution
- Use of Dot Walking for relationship mapping Final solution combines:

- Import Sets for data staging
- Transform Maps for mapping
- Dot Walking for referencing related fields

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

1. User prepares dataset (Excel/CSV)
2. Uploads data into ServiceNow
3. Data stored in Import Set table
4. Transform Map processes data
5. Data inserted into target tables
6. Relationships established using Dot Walking
7. User verifies results

3.2 Solution Requirement

Functional Requirements:

- Upload external data files
- Map fields between source and target tables
- Automate transformation
- Establish relationships

Non-Functional Requirements:

- High performance for large datasets
- Data accuracy
- Scalability
- Security

3.3 Data Flow Diagram (Explanation)

- **Input Layer:** External CSV/Excel files
- **Processing Layer:** Import Set → Transform Map
- **Output Layer:** Target tables with relationships Flow:

External Data → Import Set Table → Transform Map → Target Table → Relationship Mapping

3.4 Technology Stack

- ServiceNow Platform
- Import Sets
- Transform Maps
- JavaScript (for scripting)
- Dot Walking
- Database Tables (User, Department, Asset)

4. PROJECT DESIGN

4.1 Problem Solution Fit

The proposed system effectively addresses the identified issues by automating repetitive tasks and ensuring correct data relationships. It reduces manual intervention and increases efficiency.

4.2 Proposed Solution

Steps:

1. Upload dataset into Import Set
2. Configure Transform Map
3. Define field mappings
4. Apply scripts if needed

5. Use Dot Walking for relationships
6. Execute transformation

4.3 Solution Architecture

Architecture Layers:

1. **Data Source Layer** ○
   External files
2. **Staging Layer** ○
   Import Set
   tables
3. **Processing Layer** ○
   Transform Maps ○
   Scripts
4. **Application Layer** ○
   Target tables
5. **Relationship Layer**
   - Dot Walking

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Description	Duration
Planning	Requirement gathering	1 week
Design	Architecture & flow design	1 week
Implementation	Import sets & mapping	2 weeks
Testing	Functional testing	1 week
Documentation	Report preparation	1 week

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Functional Testing

- Verified data import accuracy
- Checked mapping correctness
- Tested relationship linking

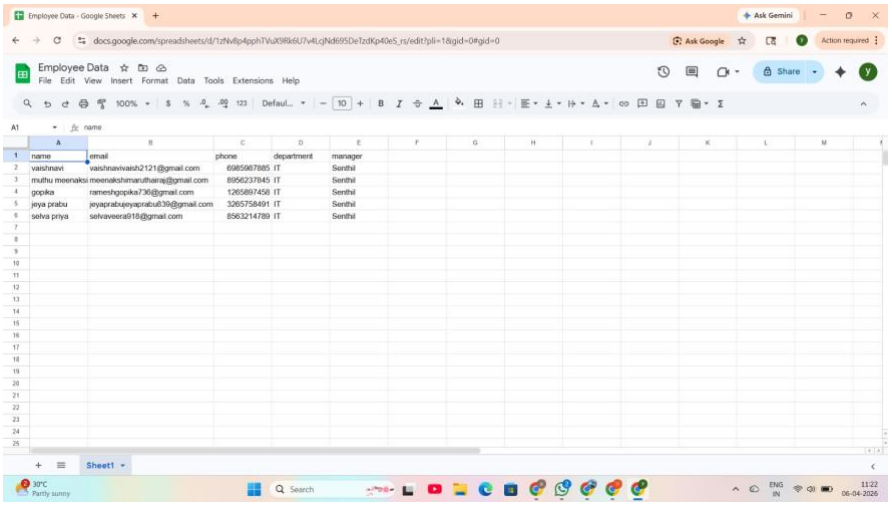
Performance Testing

- Tested with large datasets
- Measured processing time
- Ensured no data loss

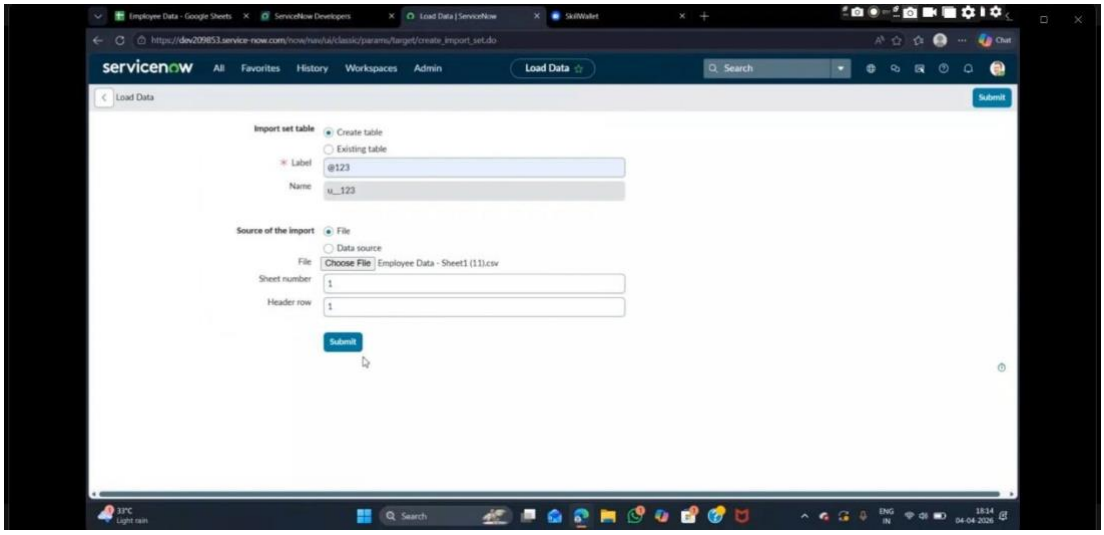
7. RESULTS

7.1 Output Screenshots

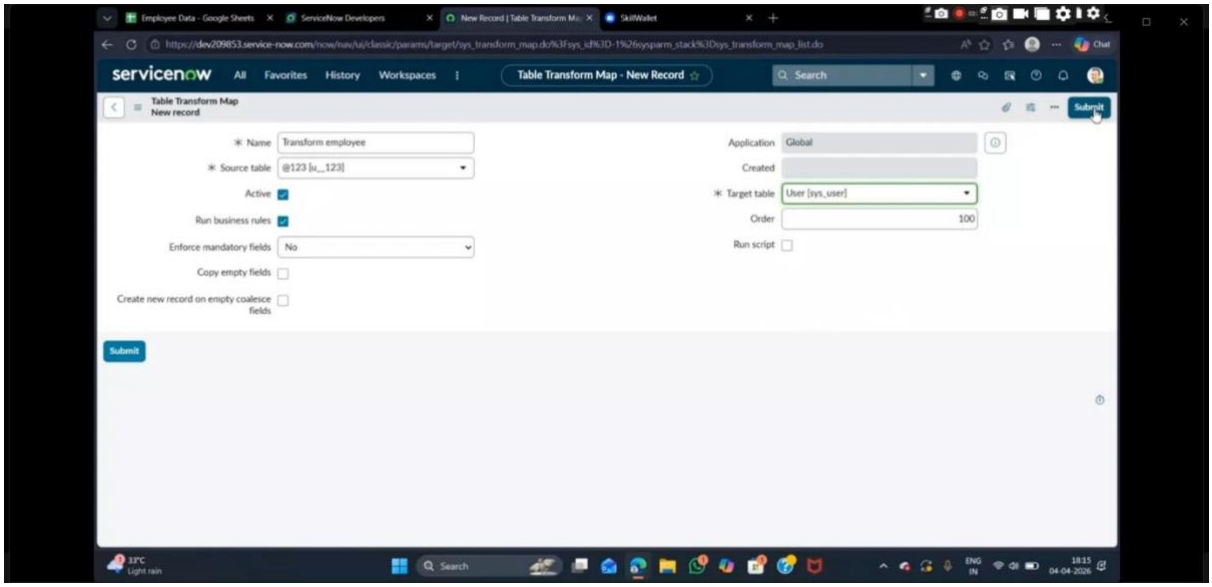
7.1.1 Create a dataset



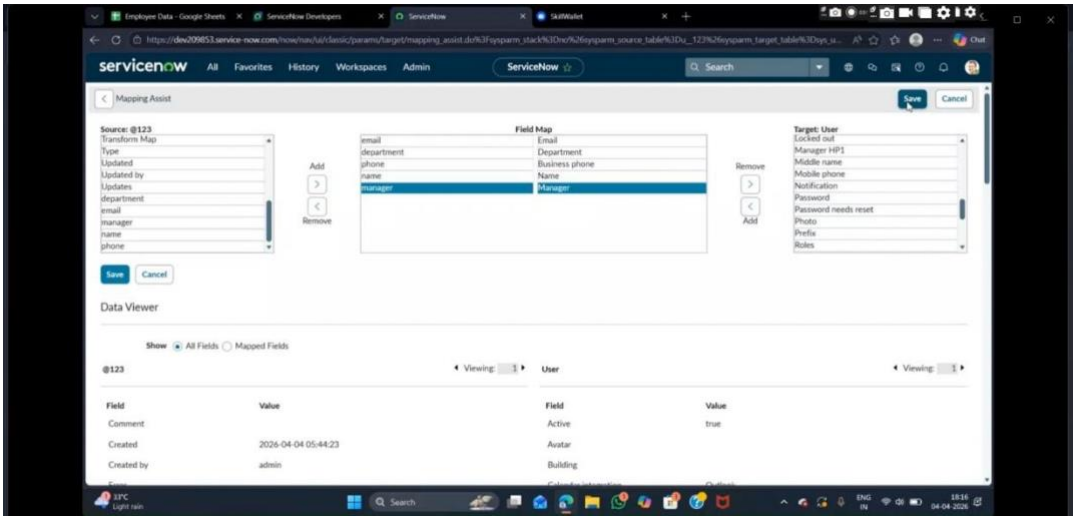
7.1.2 import the csv file into servicenow



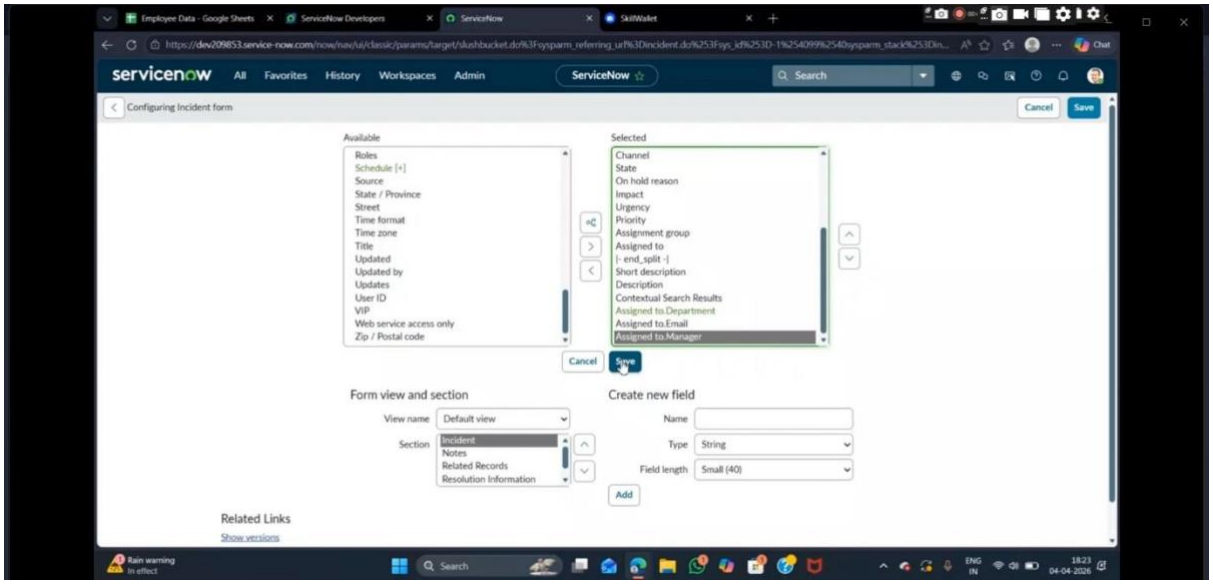
7.1.3 Create a transform map



7.1.4 Relationship mapping



7.1.5 configuring incident form



7.2 Observations

- Data imported successfully
- Relationships correctly mapped
- No duplication observed

8. ADVANTAGES & DISADVANTAGES

Advantages

- Saves time
- Reduces human error
- Scalable solution
- Improves efficiency **Disadvantages**
- Initial setup requires effort
- Requires ServiceNow knowledge
- Debugging can be complex

9. CONCLUSION

The project successfully demonstrates how automation can simplify data handling in ServiceNow. Import Sets and Dot Walking provide a powerful combination for efficient data integration and relationship mapping.

This solution enhances productivity and ensures data consistency across the system.

10. FUTURE SCOPE

The current project successfully automates data import and relationship mapping using Import Sets and Dot Walking in ServiceNow. However, there are several areas where the system can be further enhanced to improve functionality, scalability, and intelligence.

10.1 Integration with External Systems

- The system can be extended to integrate with external applications such as ERP, CRM, or HR systems.
- APIs (REST/SOAP) can be used for real-time data synchronization instead of manual file uploads.
- This will enable seamless data exchange across platforms.

10.2 Real-Time Data Import Automation

- Currently, data is imported using files (CSV/Excel).
- Future enhancements can include real-time streaming of data using web services.
- Scheduled jobs can be implemented to automate periodic imports without user intervention.

10.3 Advanced Data Validation Mechanisms

- Introduce intelligent validation rules to detect:
    - Duplicate records
    - Missing fields
    - Incorrect formats
- Implement automated error handling and reporting dashboards for better monitoring.

10.4 AI-Based Data Mapping

- Machine Learning models can be used to automatically map fields between source and target tables.
- This reduces manual configuration of Transform Maps.
- AI can also suggest relationships based on historical data patterns.

10.5 Enhanced User Interface

- Develop custom dashboards for monitoring import status.
- Provide real-time progress tracking and error logs.
- Improve usability for non-technical users.

11. APPENDIX

Dataset Link:

https://drive.google.com/file/d/1dDUTEcHyE5SeTMU2iaBobVOVUVyNpNo5/view?usp=sharing

Demo link:

https://drive.google.com/file/d/1vq9HshZfQZeVvDklBsEeejeICHKRxC29/view?usp=sharing